

Fluid Dynamics of the Atmosphere and Ocean

Problem Set 1

Unassessed.

1. Suppose that ϕ is an amount of some property of a fluid per unit volume (it is the 'concentration' of the property). Suppose that ϕ is conserved, so that $D[\phi\Delta V]/Dt = 0$. Derive the equation of motion for ϕ . Suppose that $\hat{\phi}$ measures the amount per unit mass of the property (the 'mixing ratio'). Derive the equation of motion for $\hat{\phi}$.
2. Most tracer conservation equations have a diffusion term. That is, a quantity ϕ may evolve according to an equation such as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}) = \kappa \nabla^2 \phi. \quad (1)$$

where κ is a diffusivity. In the corresponding equation for mass there is no diffusion term – it is as if $\kappa = 0$ if $\phi = \rho$. Why?

3. Using the observed value of molecular diffusion of heat in water, about $1.3 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, estimate how long it would take for a temperature anomaly to mix from the top of the ocean to the bottom, assuming that molecular diffusion alone is responsible. Comment on whether you think the real ocean has reached equilibrium after the last ice age (which ended about 10 000 years ago).
4. (a) Consider a fluid that obeys the hydrostatic relation

$$\frac{\partial p}{\partial z} = -\rho g. \quad (2)$$

Suppose also that the fluid is an isothermal ideal gas. Show that the density and pressure both diminish exponentially with height. What is the e-folding height? (This is also called the 'scale height' of the atmosphere.)

- (b) Now suppose that T depends on the northward coordinate y . Show that the eastward component of geostrophic flow is given by

$$u_g = -\frac{gz}{f_0 T} \frac{\partial T}{\partial y}. \quad (3)$$

Use reasonable values of terms on the right-hand side to estimate a value for the zonal wind in the upper atmosphere.

5. (a) The velocity in a rotating frame of reference is related to that in an inertial frame by $\mathbf{v}_R = \mathbf{v}_I + \boldsymbol{\Omega} \times \mathbf{r}$. What is the corresponding expression for vorticity? That is, how is vorticity in the rotating frame related to vorticity in the inertial frame.

- (b) Derive the equation of motion for vorticity in a rotating frame of reference. You may assume that viscosity is zero and that density is constant. (Use vectors!)
6. (a) Suppose that a sealed, insulated container consists of two compartments, and that one of them is filled with an ideal gas and the other is a vacuum. The partition separating the compartments is removed. How does the temperature of the gas change? (Answer: it stays the same. Explain.) Obtain an expression for the final potential temperature, in terms of the initial temperature of the gas and the volumes of the two compartments, and compare it to the initial potential temperature. Reconcile your answers with the first law of thermodynamics for an ideal gas, that for a reversible quasi-static process,

$$\mathrm{d}Q = T \mathrm{d}\eta = c_p \frac{T}{\theta} \mathrm{d}\theta = c_v \mathrm{d}T + p \mathrm{d}\alpha. \quad (4)$$

- (b) A dry parcel that is ascending adiabatically through the atmosphere will generally cool as it moves to lower pressure and expands, and its potential temperature stays the same. How can this be consistent with your answer to part (a)?